

A CULTURAL HISTORY OF ASTRONOMY IN JAPAN

Japan's astronomical heritage, like so much of its history, is one of enigma. Navigationally remote in ancient times and perhaps somewhat still socially remote amidst modern urbanization, it is easy to perceive that there is something unique and mysterious about what is 'Japanese'. Yet this island country is a mixture of eastern and western imports, and the social, political, and pragmatic processes related to such importation have origins dating back at least two millennia. One who actually visits the country may be somewhat overwhelmed by a large number of temples, shrines, and other architectural landmarks, some dating back many centuries, all reflecting an interaction of native socio-cultural systems with those derived from the Asian continent. Such a visitor may also be somewhat disappointed to find much of the nation's heritage engulfed by high rise buildings and the accompanying elements of industrialization, a factor which is also very much a part of modern Japan. Illumination from such industrialization is a thorn to every lover of the stars, from amateur to professional. Perhaps it is indeed the mixture of foreign imports with indigenous belief systems, a mixture more subtle than that of cultures such as the United States whose multi-ethnicity is so pronounced, that proves to be the most unique aspect of Japan. In order to understand the deeper significance of Japan's heritage with the sky, one must look beneath what appears on the surface.

For many in the West, astronomy in Japan is generally connected to Japanese names attached to comets, asteroids and supernova discovered by one of her many amateur astronomers, to astronauts accompanying NASA missions, to space probes launched from Tanegashima island, or to the building of a huge telescope on Mauna Kea sporting the name of one of the most prominent asterisms of Japanese star lore, *Subaru*. Since this volume concerns the history of astronomies across cultures, our primary focus will be on what can be termed ethnoastronomical aspects of Japan's heritage with the sky. A complete review of issues related to this heritage is far beyond the bounds of a single article. We will concentrate primarily on the cultural relations with the stars that have developed over the centuries in Japan. To provide an overview as well as a basis for understanding what astronomy means to this land, we will

concern ourselves with: (1) the development (or perhaps non-development) of astronomy as a science in Japan, and (2) the significance of history, society, and politics and their relation to star lore, mythology, and other aspects of cultural astronomy.

ASTRONOMY AS SCIENCE IN JAPAN: CHINESE AND WESTERN INFLUENCES

Many professional and amateur astronomers in Japan have asked if the ancient Japanese were not interested in the stars. (e.g., see Yokoo, 1997). Such a question reflects the fact that compared with Western traditions, it is quite difficult to find early star mythology and records of scientific or pre-scientific development that compare with many other ancient cultures. Though the exact date is uncertain, writing was introduced relatively late to the Japanese islands (the system was based on Chinese characters), and thus there is very little written record of ancient views of astronomy earlier than the 7th or 8th century.

In looking at the development of astronomy in Japan, Aveni's (1989) allusion to a kind of ethnocentrism inherent in viewing the history of astronomy only through the eyes of the modern West and its evolved science is echoed in reverse. The Meiji Reformation (1867–1868) with a de-emphasis on Buddhist traditions, followed by the post World War II rejection of the most meaningful aspects of indigenous Japanese mythology, left many in Japan with a sense that there is and always has been little which could be considered original in its own. On the other hand, the reader need only pick up any professional journal of astronomy to quickly find Japanese authors who are conducting research in astronomy using Western methodologies. Thus, to look at modern astronomy in Japan is not to look at any real cultural difference but instead to find an industrial country which is producing an increasing number of eminent Western trained scientists. The importance of understanding cultural context relative to astronomical knowledge has only recently been revived and is still not a subject for serious study in Japan.

In his *History of Japanese Astronomy*, Shigeru Nakayama (1969) gives a very thorough account of the ways in which Chinese and later Western influences were imported into Japan throughout its history. In essence, it can be seen that Japanese astronomy was greatly conditioned and restricted by geographical, historical, and cultural barriers, and its early phases were dominated by Chinese influence and later by the assimilation of Western ideas. It is doubtful that any scientific expansion of Japanese astronomy occurred prior to close contact with China or with China via Korea. Further, while the primary purpose of calendrical study in both the West and in China was to develop precise means of time reckoning, calendar development in Japan generally focused on divination and securing the position of rulers rather than trying to develop precise theoretical, cosmological, or observational methods designed to explain the workings of the universe.

Nakayama's work is written in English and parallels the two-volume work on the history of astronomy in Japan written in Japanese by the esteemed scientist and historian Toshio Watanabe (1986, 1987). A view of most Western

discussions of the subject will lead the reader to conclude that little has changed in perception relative to Nakayama's assessment (See for example Sugimoto and Swain, 1989; Pannekoek, 1961; North, 1995; Ronan, 1996; Hashimoto, 1997). While Aveni (1993) does emphasize more ethnographic and fewer ethnocentric viewpoints, he also tends to place Japan within the broader context of the 'rich Chinese heritage', and does not really deal with Japan's own astronomical heritage.

When the English scholar Basil Chamberlain was assessing the general state of affairs in Japan in his now classic *Japanese Things* (Tuttle edition, 1971), he indicated that even apart from any scientific originality, one would never find much creative lore composed in Japan relative to the stars. Viewing astronomy in Japan from only these perspectives, it is easy to get the impression that early and perhaps even later inhabitants of Japan never looked up, were never inspired by much in the sky, and patiently and/or eagerly waited for (or perhaps at times were even antagonistic toward) enlightenment from the Asian continent or later from the West, but nevertheless developed keen observational skills once such imports were implanted.

It is certainly true that almost everything which can be viewed as scientific astronomy was imported from the Asian continent or from the West in more modern times, and if only that side is viewed, discussion of the history of astronomy in Japan could basically stop here. However, to discard the culture's relations with the sky by judging it only by its development of scientific concepts dismisses what is perhaps one of the most unique case studies in how indigenous native beliefs and accompanying views of the heavens were integrated with belief systems of other cultures. Such a pattern of coexistence has been paradoxical if not down right baffling to most Western eyes. By neglecting the socio-cultural side of Japanese history, one misses the complexity of the manner in which the ancient Japanese (not only rulers, astrologers, and calendar scholars, but average farmers, fishermen, and other common citizens) incorporated astronomy into an extant system of belief.

Though we will discuss many other examples of Japanese adaptation and development in the next section, it is worth noting particular historical events that we have explicated elsewhere (Renshaw and Ihara, 1999). Consider the large influx of Chinese learning in the Asuka (late 6th to early 8th centuries AD) and Nara (710–784 AD) periods.¹ Certainly, palaces and Buddhist temples were being laid out using Chinese derived geomantic principles, and Chinese methods of calendar reckoning were adopted (Nakayama, 1969). However, each aspect of Chinese learning that was incorporated had to find juxtaposition with a set of beliefs that sometimes stood in direct contradiction. Juxtaposing the lineage of the emperor as a child of the sun goddess Amaterasu with the Chinese perception of imperial rule centered on the north celestial pole required some compromise. Buddhist temples aligned in North-South directions had to fit within a terrain that had Shinto shrines aligned in less celestial fashion but more along the flow of natural sites such as prominent sacred mountains, forests, waterfalls, or distinguished outcroppings of rocks. Besides blessing a particular Buddhist temple, a Shinto deity often had a shrine built directly

within the precincts of the temple grounds and was even given a form of Buddhahood. Compromises also included such practices as substituting Shinto ritual objects for bones of the Buddha in the treasures buried beneath pagodas (See Matsumae, 1993; Sonoda, 1993; Brown, 1993c).

The picture we see from reading Krupp's (1983, 1989) substantive accounts of temple layout and practice of seasonally based ritual in ancient Chinese capitals were only partially incorporated within the layouts and attendant rituals of ancient Japanese palaces and their imperial owners. Certainly, as will be seen in the review of social consciousness in the next section, rituals such as human sacrifice found no place in a belief system based on ritual purification and abhorrence of death. It is also significant to note that while China underwent several dynastic changes, some accompanied with revisions in cosmology and astronomical perception, Japan retained virtually one dynastic lineage reaching well into the twentieth century, thus often seeing no need for 'advances' taking place on the continent (Brown, 1993a, 1993c).

When cultural perspectives are included, it seems clear that the history of Japanese astronomy is not simply the history of Chinese astronomy or even, for better or worse, a direct mirror of such. While it may be impossible to distinguish anything as having been uniquely Japanese, cultural practices from earliest to modern times bear distinct marks of a social consciousness inherent in Japan's cultural development. Also, while Chinese astronomy may not have been given its due in modern scholarship, it is also evident that all Asian cultures cannot be lumped together with China in order to gain a full understanding of the relation between astronomy and culture in any particular one.

SOCIETY AND POLITICS AND THEIR IMPACT ON DEVELOPMENT AND ADAPTATION OF STAR LORE AND MYTHOLOGY

Ruggles and Saunders (1993) asked, 'What do people see when they look at the sky?' and responded that 'the answer is as much a cultural as an astronomical one'. Their perception is especially relevant to unraveling the enigma of Japan's heritage with the sky. A growing database of archaeological evidence, renewed post-war scholarship using historical texts, and the development of interdisciplinary approaches reveal Japan as a culture that has always been guided by an indigenous social consciousness and for most of its history a rather consistent set of social and political goals (Brown, 1993a). That was certainly a factor in how the Japanese developed myth, legend, and lore related to the stars, and it is in this area that we can find the rich sources of cultural astronomy in Japan.²

The ancient sense of social consciousness may best be understood by using the analogy of a lake fed by two springs; the lake stands for the historical and modern socio-cultural milieu of Japan, and the springs feeding it are the sources from which that milieu has been formed. The first and nearest spring is composed of a specific set of socio-political aims that we detail later. The second spring (better viewed as many secondary springs) consists of the infusion of ideas, concepts, technologies, philosophies, and religions including a virtually continuous input from the Asian continent and in later centuries large contribu-

tions from the West. The primary spring has always been fed by the second. The flow may have been slow or even imperceptible at times, but it has always been there (Oguma, 1995). On the other hand, this primary spring (regardless of its diverse historic origins) has always been deep and has formed an archetypal base through which everything must be filtered to become Japanese. At least within the last 1600 to 1800 years, the secondary spring has probably never had a direct route to Japan's archetypal lake; the flow has almost always passed through the primary spring.

Brown (1993c), Kidder (1993), and Matsumae (1993) have provided a cogent paradigm through which Japanese and attendant Shinto beliefs and practices may be understood as well as the socio-political directions that guided the early formation of Japan as an unified country. From this paradigm, three primary aspects emerge: linealism, vitalism, and optimism. Linealism emphasizes ancestral lineage with filial duty to parents, siblings, friends, nation, etc. Vitalism emphasizes life along with the abhorrence of anything that has to do with death and stresses ritualistic (not moralistic) purity. Optimism places emphasis on being concerned, not so much with the distant past or future, but rather with moving forward through seasons and cycles of life, regardless of circumstances. The socio-political goals which played a role in Japan's early development as a nation may perhaps be best paraphrased as follows: (1) unification of often warring petty kingdoms (along with their local myth, legend and lore) through (2) cooperative efforts of common people in various ways such as cultivating rice (using celestial signs and allegories for seasonal determination of planting and harvesting as well as agriculturally based festivals) in order to (3) establish a singular lineal order of imperial rule (resulting in perhaps one of the greatest national Japanese myths, that of the sun goddess Amaterasu and her place as ancestral head of the imperial line).³ It can easily be argued, given political developments of the 20th century, that while some commodities have changed and political power has not always been imperial, these fundamental aspects still form a prime keystone of the culture's psyche.⁴

One of the few writers in English to deal with Japan's cultural star lore and myth is E. C. Krupp (1991, 1997). Readers may be familiar with the several accounts he provides of the fundamental Japanese myth of Amaterasu and its relation to the Japanese political history. However, while this myth, itself closely tied with the political purpose of unification, is a prime example of lore that reflects the Japanese consciousness, there are many other Japanese traditions and bodies of star lore which show the complex relation between indigenous native belief, society, and astronomy. Little of this material has found its way into English sources, and in the remainder of this article we will discuss some of the many examples of relations between Japanese values and society and the stars. Specifically, we will look at (1) values applied to particular star groupings such as Oyaninai Boshi, (2) cooperative efforts such as planting, harvesting and fishing reflected in star lore closely tied to such asterisms as Subaru and the belt stars of Orion, (3) ancient Japanese society as reflected in star paintings of archaeological sites such as the tombs of Takamatsu Zuka and Kitora Kofun, (4) mixtures of Shinto beliefs and values seen in adaptations

of continental legends and rituals such as that of the legend of Orihime and Kengyuu and its relation to the festival of Tanabata, and (5) the pre-eminent holiday of the year for Japanese, New Year's Day, in which the values of linealism, vitalism, and optimism come together in a celestial greeting of new beginnings.

OYANINAI BOSHI: LINEALISM, VITALISM, AND OPTIMISM IN THE STARS

There are a number of celestial symbols which concern various groupings of three stars: Oyaninai Boshi including Orion's belt stars (Nojiri, 1973); three stars of the 5th moon station of the Azure Dragon of Spring, which includes Alpha Scorpio flanked by Sigma and Tau; and three stars discerned in the constellation Aquila, including Alpha flanked by Beta and Gamma Aquila (Uchida, 1973).⁵ All have to do with seeing the image of two parents standing or being supported on either side by their child in the middle. Such symbols of filial duty are numerous in Japanese star lore and reflect the strong sense of lineage later reinforced by Confucian ethics. Further, particular attention is placed on this value relative not only to parents but to friends as well. An example of a sense of Japanese identity is seen in Kenji Miyazawa's more modern lore, *The Milky Way Railroad* (Sigrist and Stroud, trans., 1995).⁶

Another legend includes the belt stars and M42 region of Orion. According to Uchida (1973), two sisters were walking down the road, the younger dutifully following her older sister and shouldering a tub of water. Being chased by an ogre, they found a rope leading to the heavens and began to climb. Though the sisters escaped the ogre, the younger sister sadly had her foot bitten off. These days, we see the bamboo pole (the three belt stars called Take no Fushi in this story) with which she continues to carry water as she follows her sister (the moon) around the sky. Her remaining foot (M42 region) peeks from the folds of her kimono. Western readers may find the end of this legend disturbing or even somewhat cruel. However, all the fundamental values of Japanese consciousness are found in this story: linealism in the form of filial duty, vitalism in strength and courage, and optimism in a will to go on to an immediate brighter future.

COOPERATIVE ACTIVITY: PRAGMATIC SIGNS IN THE STARS

The three belt stars of Orion (called Mitsu Boshi), the Hyades (Ame Furi Boshi or Rain Stars), and the Pleiades (Subaru) played a prime role as symbols of fertility and time pieces relative to seasonal change, specifically planting and harvesting during the rice cultivation period (Hirose 1972; Uchida 1973; Nojiri 1973, 1982). Their earlier and later heliacal setting in spring as well as heliacal and later rising in autumn encapsulate that season and in more ancient times provided symbols related to festivals such as spring Higan and Ura Bon. Spring Higan is celebrated within a period of seven days surrounding the spring equinox. At this time, the Pleiades, Hyades, and Orion begin to set earlier and earlier each evening. For the early Japanese, this setting was allegorical to the planting of rice seedlings (Subaru or the Pleiades appearing as united seedlings),

the rainy season (Hyades or Ame Furi Boshi seen as rain stars), and cultivation (one of the earliest symbols seen in the three belt stars of Orion was that of Karasuki or the plow). Ura Bon is celebrated in the autumn when ancestral *kami* (deities) visit and are sent on their way in thanks for bountiful harvests. In more ancient times, when this festival was celebrated on the lunar calendar based 14–16th days of the 7th month (late August or early September in the Gregorian calendar), these three prominent star patterns were seen to transit the zenith as the full moon set and dawn arrived, a signal that ancestral spirits had departed for yet another season (Miyata, 1996). The appearance of Subaru rising in the evening at this time of year seems to have had significance not only in its relation to the end of the rice harvest but to the coming of winter.

The mythology found in the *Kojiki* (*Records of Ancient Matters*; see Chamberlain, trans., 1981 or Philippi, trans., 1968) and the *Nihongi* (*Chronicles of Japan*; see Aston, trans., 1972) places much importance on the deity (sun goddess) Amaterasu, who conquered lesser deities (local states) and was considered the progenitor of the imperial line. Interestingly, one of the earliest and most significant aspects of Subaru was related to the myth of Amaterasu. Apparently taken from the phrase *mi Sumaru no Tama*, the name Sumaru was used to describe this asterism referring to the string of jewels hung on the *sakaki* tree while deities danced to lure Amaterasu from the cave (Nojiri, 1973). Allegorically, just as the sun (Amaterasu) departed for the winter, jewels (Subaru or Sumaru) appeared in the sky which could be seen throughout the winter months and served as a reminder for the early Japanese that the sun would indeed return with her spring warmth and another planting season could begin.

Lore related to Subaru is extensive, but its prime importance is that it not only played a role in what would become a central myth for unification of the country, but was also a time piece for the pragmatic needs of farmers and fishermen who were part of the unification efforts. For common people, the struggle for unification was of secondary importance. What was of prime concern was that they knew good and bad times for planting, harvesting, or setting lines for fishing. While the three star patterns we have mentioned were of significance because of their heliacal rising and setting, the three belt stars of Orion, due to their prominence, apparent equal spacing, and perceived change in position as they traced their way across the winter sky, were particularly useful.

The perceived angle of the three belt stars seen rising in the East, moving across the sky, and setting in the West at different times of the year, provided the base for farmers to use Orion as an agricultural symbol (Uchida, 1973; Hara, 1975; Nojiri, 1988). The three belt stars are variously called Awainya Boshi (Millet Stars), Komeinya Boshi (Rice Stars), or Awaine Boshi (Millet and Rice Stars). All of these names relate to seeing Mitsu Boshi as a fulcrum, balancing the yield of rice or millet crops as they move across the heavens. The star Alnilam (*Epsilon Ori*) is the center of this fulcrum. Mintaka (*Delta Ori*), being higher or lower than the center, indicates the yield of millet; Almitak (*Zeta Ori*) represents the yield of rice. In latitudes of Japan, Mitsu Boshi rises in an apparent vertical position. As the three stars move across the sky in the

fall, they appear at an angle that gives rice a strong weight on the balance; this is the time to harvest rice and plant millet. As this constellation is seen setting in the West in late spring, Zeta Ori begins to dip lower and lower; this is the time to harvest millet and plant rice.

Several symbols of agriculture, especially regarding the cultivation of rice, have been seen in the relation of the three belt stars to the M42 region. One of the oldest object references associated with this configuration uses the term *Karasuki* to describe the three belt stars as prongs of a plow with the M42 region being the handle used to pull it in the field. As mentioned earlier, this symbol is especially meaningful at the evening appearance of the three stars in the time of early spring.

As a further example of adaptation of imports to suit Japanese needs, it is interesting to look at some of the Chinese interpretations of star groupings such as Subaru or the three stars of Orion's belt and the differences in names applied by each culture. By the 7th century (and perhaps earlier), the Chinese cosmology which gave rise to the concepts of 28 *sei shuku* or lunar lodges encircling the celestial globe had found its place in Japan (Nakayama, 1969; Watanabe, 1987). The constellation of Orion contains two such *sei shuku*: the 20th which includes the small Meissa (Lamda, Phi Ori) and the 21st which embodies a much larger section of Orion and is most distinguished by the belt stars (Mitsu Boshi). Chinese characters were used by later Japanese writers to represent these configurations, however, the original Chinese term for the 21st *sei shuku* was simply *shin* (three) while earliest records of Japanese nomenclature use *Karasuki* or plow (Ozaki, 1993; Shinmura, 1994). The early Japanese gave little significance to the 20th station. Although continental influence can certainly not be dismissed as a major factor in Japan's agricultural development especially with regard to tools, it would appear that agricultural symbol in Orion may indeed have been in use before the introduction of writing and other imports of Chinese culture into Japan.

Following massive infusions of culture from China and Korea, Japanese emperors practiced the custom of plowing the first furrow around the lunar New Year. However, Japanese lore associating the three stars with a plow seems to have preceded this Chinese-based ritual.⁷ Astronomy as practiced in China during Japan's early formation as a nation had depended for some time on somewhat more precise estimates of a lunar calendar for fixing dates (Nakayama, 1969; Ho, 1985). Because of its use as an agricultural sign, especially its prominent evening setting at what later was called spring Doyou (the period of 18 days prior to the sectional term Summer Begins, the official time for preparation to begin planting rice), *Karasuki* may have originated with early Japanese farmers. Regardless, its setting has certainly survived as a symbol of planting in most rice producing parts of Japan.

Another symbol for planting indeed reflects Chinese influence, and this is clearly seen in the incorporation of *kanji* or Chinese characters. This symbol, again related to the setting of Orion in the west during the beginning of the rice cultivation season, was yet another reinforcement for use of the belt stars and the M42 region as a sign for planting. Uchida (1973) indicates that the

Japanese in some agricultural areas see the *kanji* for entry in the configuration and call it *Hoshi no Iri* (Entry of Stars). As fishermen saw stars set into the sea, Japanese farmers saw stars set into the western landscape. This symbol of *Iri* was merely another sign that it was time to plant (enter) rice seeds into the earth.

Agricultural associations discussed so far have generally been related to the setting of Mitsu Boshi during the early stages of annual rice cultivation. As mentioned, the rising of the three belt stars in the East was also an agricultural symbol and was allied with the end of the period of rice cultivation. Hara (1975) records that in many rice farming areas, the stars were called *Haza no Ma*, a term that refers to a three pole stand that is used in the field to dry rice. In the early phases of autumn, when Orion is no longer seen at sunset but rather rising earlier and earlier each night, farmers looked to the belt stars and saw them as a symbol that 'only Hasa are left in the field', the harvest of rice being over.

While the combination of a lunar calendar and later adoption of a Gregorian calendar in post-Meiji era Japan led to more precise methods for determining times for planting and harvesting, many old farmers in rural agrarian areas still use methods that are centuries if not millennia old. According to Uchida (1973), the following timepiece is still recited in such areas: 'When Mitsu Boshi are one fathom high; it's time to go to bed, when Mitsu Boshi are in the middle, it's the middle of winter; and when Mitsu Boshi lie, it's time to wake up.' This refers to the vertical alignment of the three belt stars as they rise in early autumn, their angular position in the middle of the Southern sky in winter, and their horizontal visual alignment in the West in spring. These metaphors are related to (respectively) autumn harvest, winter rest, and spring planting.

Star lore related to fishing is somewhat more rare than that related to crop

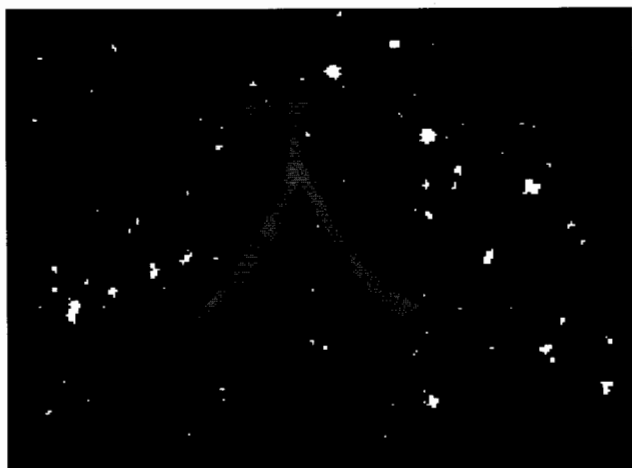


Figure 1 The *kanji* for *Iri* or Entry superimposed on the belt stars and M42 region of Orion. (Computer graphic by Steven L. Renshaw)

cultivation. It was primarily through the cooperation of local farmers along with their local ancestral *kami* or deities in the production of rice that early Japanese rulers were able to unify the country under one ruler. Still, incorporating the legends of families of fishermen was an important part of this unification, and a mix of agrarian and fishing lore is found (See Nojiri, 1982, 1987). In some fishing areas, the three belt stars are called Kanatsuki, which is a name given to a spear with three prongs used in fishing (Nojiri, 1973; Uchida, 1973). We can note some similarity in pronunciation of Karasuki (plow) and Kanatsuki. As a sign, Kanatsuki was used as a timepiece for favorable catches. When prospects seemed to be particularly good in the autumn, old fishermen were often heard to say, 'Let's wait for Kanatsuki' before going out for the evening's catch.

Perhaps one of the most significant relations to fishing found with Mitsu Boshi relates to their designation by some fishermen as Sumiyoshi Boshi. Sumiyoshi were the three deities mentioned in the *Kojiki* as being created from the ocean and being particularly favorable to seafarers (Nojiri, 1988). Incorporation of these deities in the *Kojiki* and *Nihongi* was no doubt designed to find favorable reaction from the *be* or local families of fishermen. However, their use in legends which also included allusions to agricultural symbols make them a significant part of the way in which Japanese developed celestial allegories.

The ancient Japanese were not blankly staring at their rice fields while the heavens revolved overhead. Creative and pragmatic use of star patterns played a central role in the day to day life of common citizens. Using the example of the belt stars of Orion, we have briefly discussed some of the more significant associations used in cooperative activity.

ADAPTATION OF ASIAN COSMOLOGY FOR POWER AND CENTRALITY: TAKAMATSU ZUKA AND KITORA KOFUN

Some archaeological sites may provide cogent information about the early interaction of Japanese values with imported cosmology. One of the major archaeological discoveries was made in the early 1970s (Hirose, 1972) in Asuka village, Nara prefecture. From an Edo Era painting showing this mound with a tall pine tree atop, this tomb was called Takamatsu Zuka Kofun (Tall Pine Burial Tomb).

Dating indicates that the tomb was built in the latter part of the 7th or early part of the 8th century. As was and is the case with most burials in China, Korea, and Japan, the tomb was aligned with celestial north. Paintings of animals related to the four cardinal directions were found on the walls, and careful inspection of the ceiling revealed a chart including the 28 *sei shuku*. Only a few scholars knew of the tomb's existence in the Edo era (1603-1867) and most believed it was that of the emperor Monmu (r. 697 to 707). Modern excavation revealed no inscription, and Monmu's tomb was later determined to be to the East. Similar tombs have been found in both China and Korea, and the construction period was also one in which scholars from Korea had been invited to the imperial court.



Figure 2 Takamatsu Zuka Kofun is aligned with celestial north. This view from due south shows the bamboo covered mound of the tomb. The doors are to chambers of relatively recent origin; inside are housed temperature maintenance and dehumidifying equipment used to protect the small tomb's delicate paintings. The tomb is not opened to the public. (Photo by Steven L. Renshaw)

What is significant about Takamatsu Zuka Kofun is that it clearly shows the influence of Chinese and Korean cosmology on Japan in the Asuka Era (late 6th to early 8th centuries). In 1998, another such tomb, located about 1 km to the south of Takamatsu Zuka Kofun on Mount Abe and named Kitora Kofun (after the Kitaura area of Asuka village), was explored. The tomb was not actually entered but probed with a sub-miniature camera. It was dated within the same Asuka period as Takamatsu Zuka Kofun. While there are some remarkable similarities, there are also some anomalous differences in the paintings of the two tombs. (See Inokuma, Izumimori, Kawakami, Sawada, et al, 1999). In Kitora Kofun, the paintings of the animals of cardinal directions are in somewhat better condition than those of Takamatsu Zuka Kofun, in which some paintings appear to have been defaced in ancient times. The animals were painted in a freer style than those found in the tomb to the north. There also are more stars in the Kitora tomb paintings of the sei shuku as well as many other constellations not found in the Takamatsu Zuka tomb. Unlike Takamatsu Zuka Kofun, circles representing circumpolar stars, the celestial



Figure 3 Viewed from South to North, Kitora Kofun (Kitora Tomb) as it appears today in Kaitaura Village, Asuka, Japan. The coffin, which was explored using a sub-miniature camera, is located in the lower mound and is aligned with due north. (Photo by Steven Renshaw)

equator, the ecliptic, and the observer's horizon are distinct. Like Takamatsu Zuka Kofun, the tomb seems to have been plundered of any treasure it may have held. The south wall of Takamatsu Zuka Kofun was obliterated by an entryway in ancient times, thus eliminating the painting of the Red Bird of Summer. Because of limitations in the flexibility of the sub-miniature camera, the existence of this symbol on the south wall of Kitora Kofun is not yet confirmed.

With regard to astronomical content, there are several significant findings (Miyajima, 1999a; 1999b). First, the paintings of sei shuku and other constellations in Kitora Kofun are based on different Chinese/Korean charts from those of Takamatsu Zuka Kofun. They use a circular configuration rather than the square one found in the latter tomb. Second, the point of the crossing of the ecliptic with the celestial equator can be seen in Kitora Kofun and is near a point in Aries closer to Taurus and corresponding with a position several centuries earlier than the Asuka era or the tomb's construction (the vernal equinox was shifting into Pisces by the end of the 7th century). Third, since the ratio of the northern inner circle depicting stars which were seen not to set can be proportionally matched with the angular measure of the chart observer's latitude and the celestial equator, a relatively good estimate of observer latitude can be made. Fourth, the painting of the White Tiger of the West is reversed from that of Takamatsu Zuka Kofun and tombs with similar paintings found in China. In Kitora Kofun, the tiger is painted facing north. When looking at the west wall of Takamatsu Zuka Kofun,⁸ the tiger is seen to face south, and this indeed corresponds with the placement of the figure of this animal among the stars.

Miyajima's research indicates that the chart was based on observer data gathered most probably around 65 BC. It was adapted by an astronomer who lived in a more northern portion of East Asia such as the ancient Korean kingdom of Kokuryo (around present day Pyongyang), and then was brought into the Asuka area much later. It appears that the chart's painter in Japan lacked full understanding of the underlying astronomical principles used in the chart's original observational data and construction.

These tombs provide us with a view of how Chinese/Korean astronomical thought found its way to Japan, and how such thought was incorporated into the social programs of Japan's rulers. It is not the accuracy of the cosmology reflected in the tombs' paintings but rather the almost wholesale adoption of Chinese/Korean ideas to further strengthen the centrality of imperial rule that was most important, especially in the case of Kitora Kofun. Japanese rulers wished to give themselves the same legitimacy that Chinese rulers had; incorporation of symbols and astronomical methods in order to accomplish this did not necessitate complete awareness of the underlying principles, but only their mere presence. As mentioned earlier, such was the case with the introduction of Buddhism as well as other aspects of Chinese thinking. One can still see this use of foreign imports without substantive understanding in the almost playful use of foreign languages in what sometimes seems to be gibberish in slogans and advertising in modern Japan. Use of foreign icons brings esteem, regardless

of whether or not they are understood. Again, Japanese beliefs and rituals have remained somewhat constant regardless of foreign influences.

ADAPTING CHINESE LORE TO NATIVE BELIEFS: ORIHIME, KENGYUU,
AND TANABATA

When we look at the adaptation of myth and legend imported from other cultures, we also find the sense of indigenous Japanese values. Perhaps one of the best examples of this is the legend of Orihime and Kengyuu (Nojiri, 1973). This story and its associated festival were probably imported from China in the Heian Era (794–1185). The story involves the stars of Vega and Altair, and the reader should consult Krupp (1991) for an explanation of the story in its Chinese form. Essentially the same in character, there are some noticeable adaptations made. In Japan, the star Vega is often called Orihime Boshi (Weaving Princess Star), and Altair is often called Kengyuu Boshi or Hiko Boshi (Puller of Cows Star). To give the reader one Japanese version of the legend, we will paraphrase Hara (1975):

One day, the emperor's daughter, Orihime, was sitting beside the Milky Way. She had been weaving because her father wished it, since he loved the beautiful clothes that she made. On this day, she was very sad because she realized that she had been so busy that she didn't have time to fall in love. Her father, Tentei, the ruler of the heavens, felt sorry for her and arranged a marriage with Kengyuu, who lived across the river, the Milky Way. Their marriage was one of sweetness and happiness from the start, and everyday thereafter they grew happier and happier. But Tentei became very angry, because in spending so much time in her happy marriage, Orihime was neglecting her weaving. Tentei decided to separate the couple, so he placed them back in their original places, separated by the Milky Way. On only one night of the year would he allow them to meet, the 7th day of the 7th month. Every year on that day, from the mouth of the river (the Milky Way), the boatman of the moon comes to ferry Orihime over to her beloved Kengyuu. But if Orihime has not done her weaving to the best of her skills and ability, Tentei may make it rain. When it rains, the boatman will not come, because the river is flooded. However, in such a case, Kasasagi (a group of magpies) may still fly to the Milky Way to make a bridge for Orihime to cross.

Related to this legend, the ancient Japanese celebrated the festival of Tanabata on the 7th day of the 7th month each year (lunar calendar). The 7th day of the 7th month generally falls in August or September in the Gregorian calendar. At this time of year, the constellations of Lyra and Aquila are prominent in the evening sky with their major stars, Vega and Altair, separated by the Milky Way. The 7th day of the 7th month also finds a waxing crescent moon (boat) reaching its first quarter. If it is not raining, both Orihime Boshi (Vega) and Kengyuu (Altair) are quite conspicuous at the time of the Tanabata festival.

Tanabata may be translated as 'weaving with the loom (*bata*) placed on the shelf (*tana*)', and the festival celebrates the improvement of technical skills. As in China, the ancient Japanese added specific values to their wishes that Orihime hone her skills and work hard so that she could meet Kengyuu. In modern celebrations of Tanabata, people throughout Japan write wishes (generally for themselves or relatives) to the *kami* (deity) Orihime on colorful strips of paper. On the evening of Tanabata, they tie these paper wishes to freshly

cut bamboo. Wishes may be for increased skills in work or school, but may also be for anything that reflects a person's dreams and hopes for the future. Summer vegetables such as eggplant and cucumbers are prepared, and horse or cow figures made out of straw and water oats are decorated. While the myth probably held seasonal significance in its Chinese origins, specifically the celebration of the end of the rainy season (reflected in a desire that it not rain), it found a variety of interpretations related to seasonality in its Japanese form. Particularly in relation to agricultural development in Japan, wishes related to celebrations of Tanabata ranged from desire for dry weather to desire for wet weather depending on the particular geographic region and whether a crop was to be planted or harvested at this time.

Following Shinto practice, the concept of purification (generally including the use of water) before the Bon festival (centered on the 15th day of the 7th month) was also added to the Tanabata festival. Before the legend was brought from China, a festival had been held to welcome the water *kami* at this time of year; infusion of the legend of Orihime and Kengyuu added a motif of the marriage of a weaving lady and the water god (Okada and Akune, 1993). In eastern parts of Japan, an associated ritual called Nebuta was celebrated. On the early morning of Tanabata, bamboo would be set afloat in the river, and people would brush their bodies with leaves from silk trees. By doing so, they were said to take their sleepiness (*nebuta*) away, another form of purification and preparation for Bon (Yoshinari, 1996). The close relation of Tanabata to the indigenous Bon festival has led to a number of adaptations of Chinese mythology. In short, one makes the coming of the Bon festival sacred by excluding impure spirits from the body at the first quarter moon, thus insuring purity for the coming of Bon at full moon. It is interesting that in some regions of Japan, Tanabata is accompanied by a taboo forbidding swimming or bathing in a river. Noting the relation with the celestial river or Milky Way, the taboo is based on the idea that a Kappa or water deity resides in the river, and one should not make the water dirty by entering the water deity's home.

When it was first imported, Tanabata was celebrated only by imperial court officials. It was considered a graceful event, full of the simple elegance associated with the Heian era. Lanterns were lit and poems were written on mulberry leaves still holding their dew (Nojiri, 1973). As the custom spread to local areas, towns became covered with bamboo at Tanabata, and the festival took on more of the Japanese values.

The process of adapting an imported legend and developing indigenous practices evolved in complex ways over the centuries, and we have touched on but a bit of this complexity. In modern times, the festival is generally celebrated on a solar July 7th, a date that is generally still within the rainy season. Sadly, the festival has lost much of its seasonal significance with modern industrialization. Of course, the ethic of improved work and skill is still valued, regardless of whether or not the day of celebration is attuned astronomically.

PREVALENCE OF THE SUN: AKEMASHITE OMEDETOU GOZAIMASU

No discussion of Japan's connection with the sky can avoid the significance of the sun. Symbolism related to the sun is heavily incorporated in Japanese myth



Figure 4 Decorated Strip of Tanabata Bamboo from the Tottori Region of Japan. Note the representation of Hiko Boshi (Altair and two flanked stars in Aquila) at the top and Orihime Boshi (Vega flanked by two stars in Lyra) near the middle. From Nojiri, Houei. *Nihon Seimei Jiten* (Dictionary of Star Names in Japan), 1973, p. 60. Used with the kind permission of the publisher.

and is still seen in its flag. Being the most eastern of East Asian nations and indeed being seen as the land which the sun first greets in the morning, the phrase 'land of the rising sun' developed special meaning for the Japanese from ancient times. Along with the significance of Amaterasu as sun goddess, there are other ways in which the sun has become a central part of Japanese consciousness. The celebration of the New Year represents yet another way in which indigenous values are mixed with Chinese imports to provide a kind of unique cultural perspective on the significance of vitalism and optimism.

While Christmas is celebrated in Japan to some degree, it is not considered a national holiday, and the attendant Western religious aspects of that day are certainly missing. New Year's Day is by far the more significant holiday. Before the Meiji Restoration, the New Year was celebrated according to the Chinese lunar calendar. In modern times, although the lunar calendar still has influence on scheduling festivals and celebrations, the Gregorian calendar change is celebrated by most people as the official New Year. In the days before the New Year begins, people busily prepare by cleaning house and cooking food to

welcome the *kami* of new life. Rail and air terminals are jammed with people trying to get back to their home towns to spend the New Year's night and daybreak with family members.

Japanese express wishes for the New Year by saying 'Akemashite Omedetou Gozaimasu'. Only one *kanji* is found in this phrase, within the first word. This is a combination of the characters for sun and moon, and among other meanings, entails the sun and the moon getting together and becoming bright. It represents changing and opening – in a sense, dawning.

In ancient lore, under the lunar calendar, the New Year was seen in relation to change in both the sun and moon as well as in the symbolism of their luminance. The meaning of the phrase 'Akemashite Omedetou Gozaimasu' reflects central Japanese values. Perhaps it is expressed in English most appropriately as follows: 'The year is changing; darkness gives way to light; new life begins; Congratulations!' Following tradition, many Japanese on New Year's morning brave the cold to find places with unobstructed views of the eastern horizon and await the rising sun. Incorporating the ideas of much ancient mythology, the sun is seen to be making its journey back to the North, and the Vernal equinox is eagerly awaited.

As in other aspects of Japanese astronomy, Chinese imports and traditions were incorporated from early times. The New Year of 1996 was special in that it began another 12 year cycle of the Chinese calendar, based on the positions of Jupiter with its 12 year orbit and its consequent position about the ecliptic (Uchida, 1981). The tradition of using Chinese based animal names for the 12 directions and associated years is popularly maintained in Japan. In 1996, things turned around once more to the direction of *ne* (mouse), the North, to the direction of the star Polaris, sometimes called *Ne no Hoshi* (mouse star)



Figure 5 The kanji for *Ake*. Symbols for the sun and moon are placed together in this character to represent 'bright' or 'beginning'. (Computer Graphic by Steven L. Renshaw)

but also called Shin Boshi, the Heart star, the soul of the Heavens. 1996 began the clockwise cycle again which from ancient Chinese geomancy means moving from N to NNE (1997, *ushi* or cow), to ENE (1998, *tora* or tiger), to E (1999, *u* or rabbit) and so forth.

The mixture of Chinese geomancy and symbolism with indigenous values is clearly seen in the Japanese celebration of the New Year. Yet the quiet tolling of temple bells on New Year's Eve brings a certain calm and solitude unseen in many other cultures. It is in this very symbolic act that centuries of Japanese history and tradition can be seen and truly appreciated.

HOPES FOR THE FUTURE BUILT ON A RICH BUT HIDDEN PAST

Most of Japan's recorded history reflects a culture continually influenced by imports, even when on the surface the country appeared to be isolated. At the same time, a cultural consciousness combined with quite distinct social and political structures has always been a source through which most ideas, foreign or domestic, had to be filtered.

We can still see much of the ancient heritage of Japanese astronomy in layouts, structures, mythology, and ethnographic lore. As their culture has adopted almost every modern concept that East or West has to offer, many Japanese themselves are unaware of the richness of their ethnoastronomic heritage. As in the West, many young Japanese are more likely to know the name of Pleiades rather than Subaru and associate the latter with an automobile company rather than the celestial symbol which meant so much to their ancestors. While Tanabata is still observed, and the legend of Orihime and Kengyuu still finds its way into modern songs and prose, few people are concerned about the festival's shift away from its seasonal association. In many ways, however, Japan's embracing of the West and virtual denigration of its past reflects a process which has been a part of the culture's way of handling infusion from its earliest times.

Aveni (1989) alludes to the modern inclination to disassociate science from the social and historical context in which it was developed. He points out how such isolation lacks the notion of process or change and says that in a sense, we 'seek ourselves in the tattered walls of others cultures'. Aveni further mentions the danger of relying solely on written records and emphasizes using additional evidence in the form of archaeology and architectural icons to understand a nation's cultural astronomy. In taking a narrow view of astronomy as it developed into modern science, he contends that we pursue what is interesting to us instead of what was important to the people we study.

The specific irony of Aveni's observations relative to the study of astronomy in Japan may rest in the view taken by many Japanese. In a sense, the view is that we (Japanese) seek others' selves (Western science in particular) within the tattered walls of our own culture and come up sadly lacking. To further paraphrase, we (Japanese) tend to pursue what is of modern interest (to the West in particular) instead of what was actually important to our own ancestors throughout history. These perhaps enigmatic Japanese views are the result of

a long and complex history involving interaction of native belief and consciousness with a perception that while unique, the native culture is somehow inferior to those imports it filters. Such lack of cultural self-esteem has led to a number of enigmatic events in Japanese history, not merely the denigration of its own astronomic heritage.

Modern astronomical practice, while totally within the rubric of Western science, reflects a long heritage of curiosity combined with concern for future generations. A good example of this may be found in the diary of a young Kochi resident which was rediscovered in the 1980s. In 1664, while astronomers in Europe were trying to learn more about the motion and nature of comets, Matasaburou, a 12 year old boy, began to observe what would later be known as Comet C/1664 W1. Encouraged by his teacher, Jian, he diligently observed the comet for over four months, wrote his impressions, and drew the changing shape of the comet in his diary. Despite the lack of astronomical knowledge, Matasaburou showed remarkable curiosity about the true nature of the *houki boshi* (brush star) and an abundant skepticism of the prevailing view among townspeople that the comet signaled doom. While his drawings do not have the flamboyance of those of Hevelius and others observing the comet in Europe, his diary shows every attempt to plot the location and shape of the comet precisely as it crossed the celestial sphere nightly. Matasaburou passed on to his own pupils the joy of discovery, and it is no doubt due to this that copies of his diary have survived the centuries. Such diligence is still seen in the work of amateur and professional astronomers alike.

Future astronomers, archaeoastronomers and ethnoastronomers in Japan will no doubt come from an increasing number of young Japanese who have grown up in the intellectually free environment of modern Japan. Those not raised within the culture may find the door to Japan's astronomical past opening wider if there is a will to engage in an exploration of consciousness still so evident beneath the surface of Japan's modern society. However, much work lies ahead in uncovering the vast richness of Japanese astronomy in history, and the future is probably most optimistic for those young Japanese scholars who find the riches of their heritage and culture inspiration and justification enough to seek knowledge about their nation's long interaction with the sky.

NOTES

¹ An exhaustive account of these periods and the influence of continental imports is far beyond the scope of this article. The reader may find articles from Volume I of *The Cambridge History of Japan* series including Brown (1993a, 1993b, 1993c), Kidder (1993), Inoue (1993), Naoki, (1993), Matsumae, (1993), and Sonoda (1993) particularly relevant to a deeper understanding of these processes.

² A good chronology in English of Japan's general history may be found in Torao and Brown's (1991) *Chronology of Japan*. The most exhaustive history of Japan in English is the recently released Cambridge series (Hall, Jansen, Kanai, and Twitchett *et al.* eds., 1993 and following). For shorter and perhaps more approachable views of Japanese history written in English, the following may be of value: Sansom's (Tuttle Edition, 1974) pre-war but still useful 3-volume *A History of Japan* and his (Tuttle Edition, 1973) *Japan: A Short Cultural History*; Hall (Tuttle edition, 1971) *Japan From*

Prehistory to Modern Times; Mason and Caiger (Tuttle Edition, 1973) *A History of Japan*; and Morton (1994) *Japan: Its History and Culture*.

³ From our earlier discussion, the reader should have a good idea of the kinds of conflicts many imported ideas created for early Japanese. Other brief examples will further clarify the point: while the infusion of Confucianism reinforced many aspects of 'lineality', especially with regard to the development of bureaucratic governmental practices, the Confucian idea of a cyclical rise and fall of dynasties was never really accepted in Japan and stood directly against the idea of a singular imperial line. The 4th and 5th century infusions of Buddhism, with beliefs which emphasized a better life in death than in the present, directly contradicted the concept of 'vitalism' and 'optimism'. Also, the Buddhist idea of eventual decay as well as the Confucian idea of a 'glorious past' were incongruent with virtually all of Japanese beliefs.

⁴ It is easy for the reader to get full insight into the significance of these ideas by reading the many excerpts of manuscripts from periods throughout Japan's history which are accompanied by commentary in the excellent *Sources of Japanese Tradition* compiled by Tsunoda, deBary, and Keene (1964). Shinto beliefs and their pervasiveness in modern Japanese life are also discussed by Shigemitsu (1996).

⁵ Here and in later sections we discuss the Chinese derived *Sei Shuku* or 'moon stations' along with the animals of cardinal directions. Though used extensively in Japan from earliest Chinese infusions, a discussion of their complexity is more appropriate in a work on Chinese astronomy. The reader is urged to look at sources such as Needham (1959), Ho (1985), or Nivison (1989). (Editor's note: please see the Chapter by Sun Xiaochun in this volume.)

⁶ *Douwa Shu; Ginga Tetsudou no Yoru*. (A Collection of Tales; Night of the Milky Way Railroad) was originally started in 1924, adapted for some years, and finally published posthumously in 1951. It remains one of the most popular stories in Japan, loved by children and adults.

⁷ See Krupp (1983, 1989) for a description in English of this and other cultivation related ceremonies that were practiced by the emperor in China. As they were recognized in Japan, some became part of the imperial ceremony, but many never found a place either because they did not fit within the imperial goals mentioned earlier or because they were in direct conflict with native Shinto belief (ritual sacrifice, for example).

⁸ Note the photograph of the White Tiger in the Chinese tomb of similar dating in E. C. Krupp's *Echoes of the Ancient Skies*, 1983, p.112.

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